# MICROECONOMICS

MICRO-08

**Time:** 5 minutes

**Outcome:** Connect Elasticity and Total Revenue.

**Difficulty:** Intermediate

# Elasticity and Total Revenue

## THE CORE IDEA

Total revenue equals price multiplied by quantity, and the effect of a price change depends on demand elasticity. A point on demand supplies the price and quantity needed to calculate revenue. At unit elasticity the price and quantity percentage effects offset, marking the revenue peak on a standard linear demand curve.

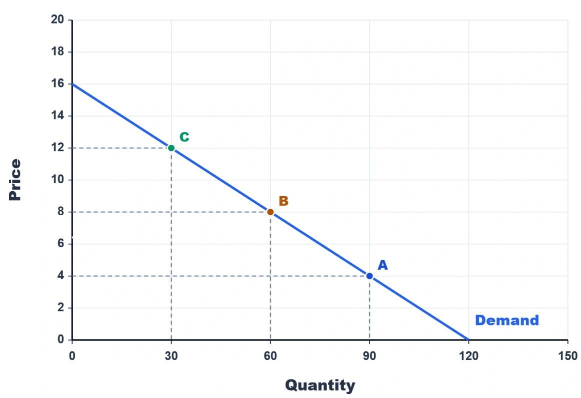
## HOW TO RECOGNIZE IT

- Calculate total revenue as price multiplied by quantity at each relevant point.
- If price and total revenue move in opposite directions, demand is elastic over that range.
- If price and total revenue move in the same direction, demand is inelastic over that range.
- At unit elasticity, a small price change leaves total revenue approximately unchanged.

## WATCH OUT

A common mistake is to compute elasticity without checking the associated total-revenue change. With elastic demand, price and total revenue move in opposite directions; with inelastic demand, they move together.

## WORKED EXAMPLE: ELASTICITY AND REVENUE




At point C, price is $12 and quantity is 30, so total revenue is $12 x 30 = $360. At point B, price is $8 and quantity is 60, so revenue is $480. Moving from C to B lowers price but raises total revenue by $120, indicating elastic demand over that segment.

## CHECK YOURSELF

For luxury cruise cabins, price changes from $3200 to $2880 while quantity demanded changes from 1,500 to 1,900. Use the midpoint method and the observed revenue totals. Which statement is correct?

**READY?** Return to the game and master this concept.